



FEBRUARY 8–12, 2027

IAS WORKSHOP

Information Theory, AI, and Language

Language was long thought to be uniquely human. Systems trained to predict the next word now infer, abstract, and reason. They are the first fully functional statistical models of natural language, and we still have no principled account of why they work. This workshop pursues a mechanistic understanding of language and language models.

FIRST PRINCIPLES

Theory-grounded models of language and language models.

EMERGENT FUNCTION

How intelligence arises from next-word prediction.

BRIDGING SCALES

From neural computation to linguistic structure and to cognition.

CONVENING

Physicists · Neuroscientists · Mathematicians · Computer Scientists · Cognitive Scientists · Linguists

ORGANIZERS

Weishun Zhong

Misha Tsodyks

Uri Hasson

Sebastian Seung